The offset is an index that indicates the position of a record in a partition and is stored with the record, as you saw in the previous example. Each consumer—defined by its `groupId`—can store the offset of the last element read in Kafka or elsewhere.

Each time a consumer starts reading a topic, you can specify the offset from which the consumer begins reading records. Some of those choices are to start reading from the first or the last item.

The *committed position* refers to the offset relative to the consumer, which is stored in permanent memory so it can be retrieved in the event of a failure. In this way, a consumer can resume reading if communication is interrupted, whether the customer gives up or the broker fails.

Consumer groups and topic subscriptions

A consumer group is a single logical subscriber that can contain multiple consumers that share the consumption of one or more topics. The pairing between the partitions of the topic and the consumers can dynamically adjust to the increase or decrease of the consumers. This is called rebalancing. Depending on the ratio of partitions and consumers, the partitions are assigned to consumers, as shown in the following figure.